

18-320 Final Project

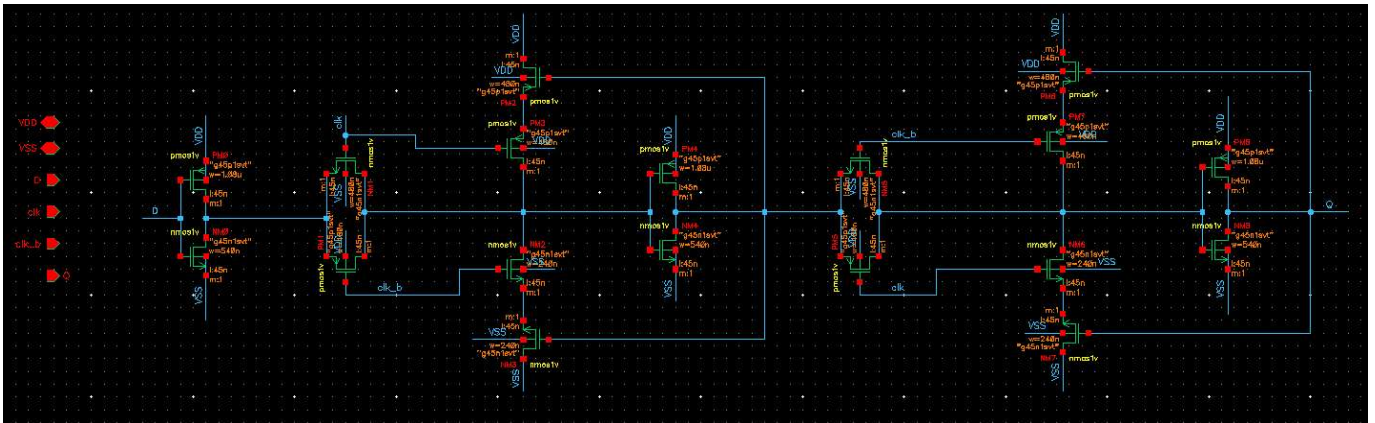
Module 1: Flip-Flop Schematic & Testbench

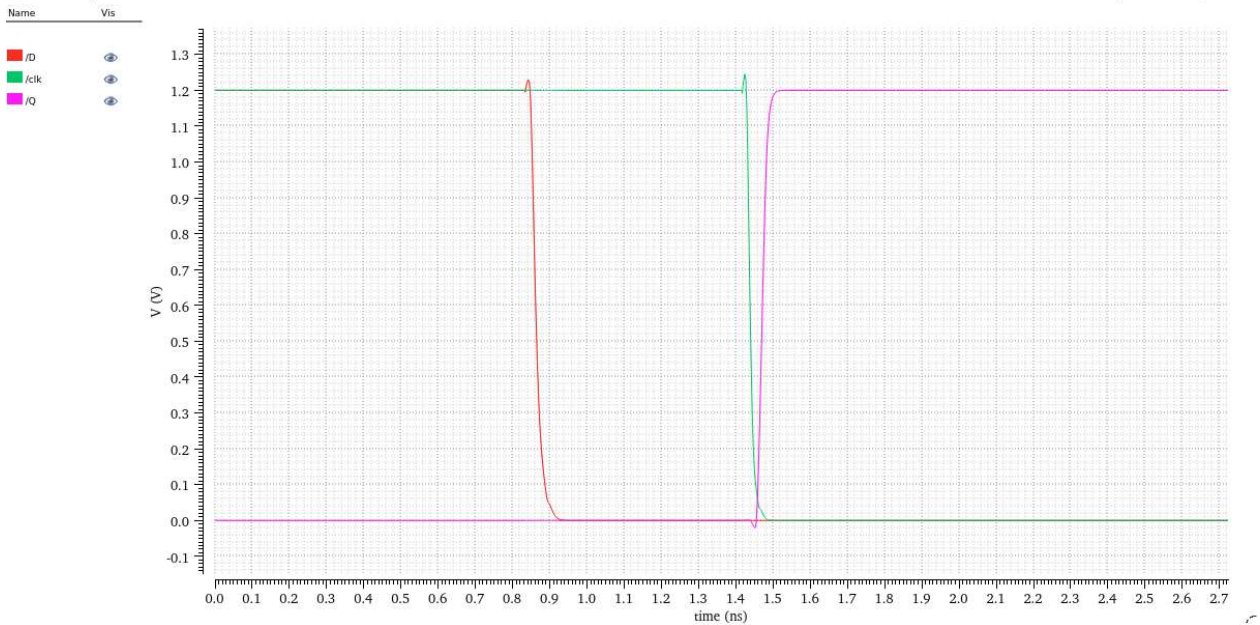
A standard latch schematic contains an inverter and a transmission gate, followed by a pair of back-to-back inverters that act as a latch. However, a flip-flop is built as a pair of back-to-back latches.

The MSFF schematic operates as a series of two separate latches, with the first latch being the master, and the second latch being the slave. When the clock signal transitions from low to high, the master latch stores the input signal. When the clock signal transitions from high to low, the signal from the master latch then gets stored in the slave latch. This is why the flip-flop is edge triggered.

Adding an extra buffer to the load improves the integrity of the output signal, as it takes time from the rising edge of the clock for the output to be seen. In addition, the buffer can reduce the capacitive load of the next stage on the flip-flop, minimizing the delay.

Note: This flip-flop is inverting and negedge triggered.





Maximum value of delay_in that you can use before the flip-flop no longer works: 544 ps

Setup Time Estimation:

Midpoint of CLK transition: 1.438 ns

Midpoint of D transition: 1.406 ns

$$t_{CLK} - t_D = 1.438 \text{ ns} - 1.406 \text{ ns} = 32 \text{ ps}$$

$$t_{SETUP} = 32 \text{ ps}$$

The setup time denotes the minimum amount of time needed that D must be held before the clock edge in order for Q to respond. In a larger circuit with multiple stages, there may be multiple flip-flops, each of which have inputs that rely on earlier stages. If the output of earlier stages takes too long to propagate, then the signal will not reach the flip-flops in later stages fast enough to satisfy setup time, creating errors. For example, as we saw in the simulation, the flip-flop output Q did not rise once the input delay was greater than 544 ps. In a real world scenario, all inputs would need to reach this flip flop within 544 ps from the clock edge or faster to be registered.

Module 2: Datapath Schematic & Testbench

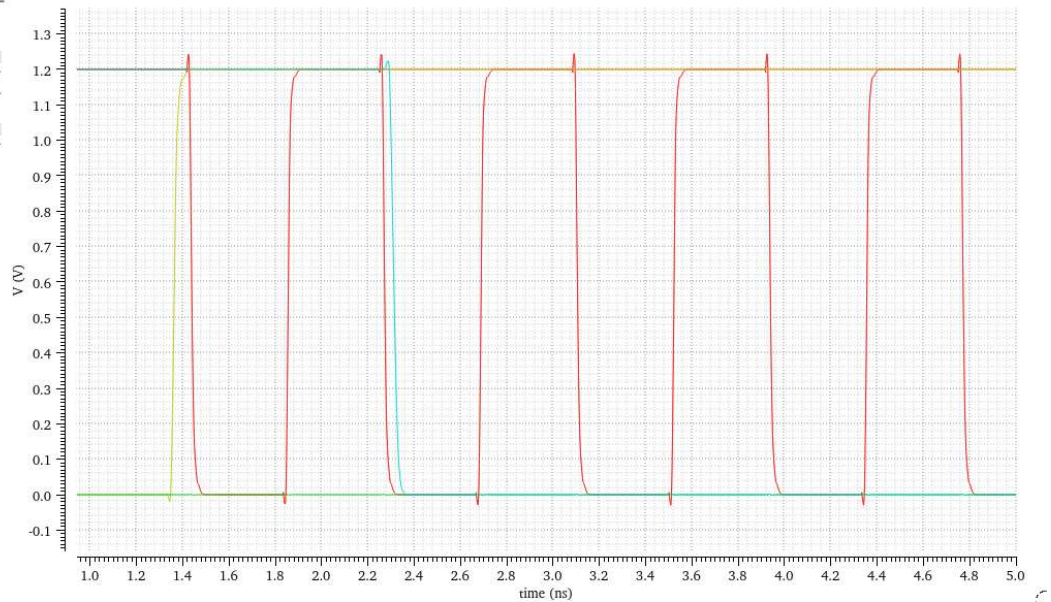
This datapath operates by placing inputs into the first set of flip-flops, then performing a series of combinational logic operations until the output is stored in the second set of flip flops. Since the flip-flops are inverting, the purpose of the inverters after the first set of flip-flops and after the second set of flip-flops is to “undo” this inversion. We can describe the relationship between the inputs and outputs at any stage with a table, then turn this into some simple equations:

in[1]	in[0]	out[1]	out[0]
0	0	1	1
0	1	0	1
1	0	0	1
1	1	0	0

$$\text{out}[1] = \sim\text{in}[1] * \sim\text{in}[0]$$

$$\text{out}[0] = \sim(\text{in}[1] * \text{in}[0])$$

Name	bit
/clk	
/D_IN<1:0>	
/D_IN<1>	
/D_IN<0>	
/D_OUT<1:0>	
/D_OUT<1>	



Combinational logic delay:

$$t_{COMB} = 45.24 \text{ ps}$$

Calculating setup slack:

$$t_{CLK} = \frac{1}{2.4\text{GHz}} = 416.7 \text{ ps}$$

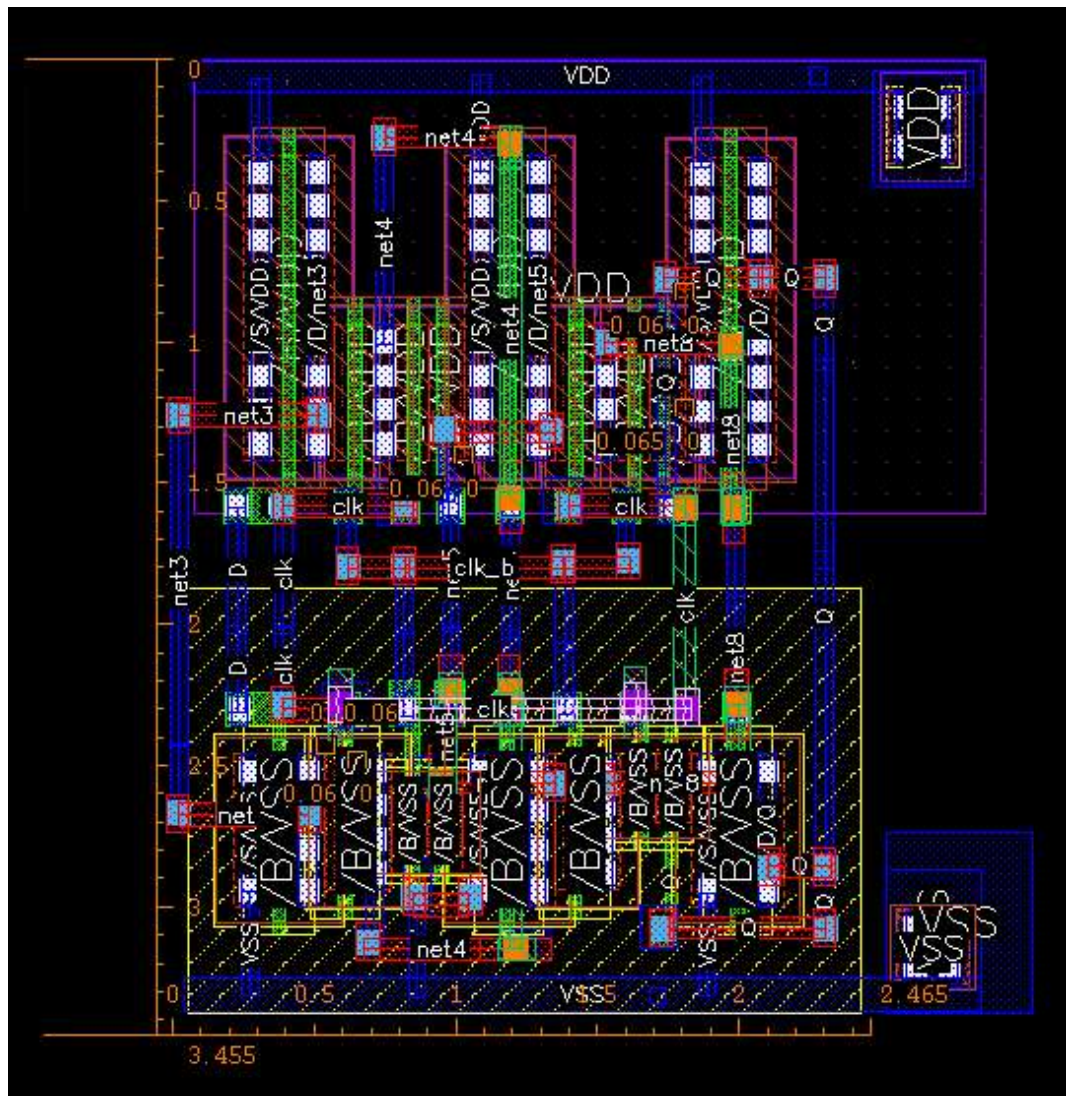
$$t_{REQUIRED} = t_{CLK} - t_{SETUP} = 384.7 \text{ ps}$$

$$t_{SLACK} = t_{REQUIRED} - t_{COMB}$$

$$t_{SLACK} = 339.46 \text{ ps}$$

Based on my calculated setup slack, the data has around 307 ps of extra time to spare after reaching the second flip flop before a setup time violation would be created. However, if we started to see a significantly faster clock, then a setup time violation could happen. Specifically, this would happen as soon as the period of the clock became equal to the setup time added to combinational logic time. One design change that could be used to improve the timing margin would be to create faster combinational logic gates, which would further increase the slack time.

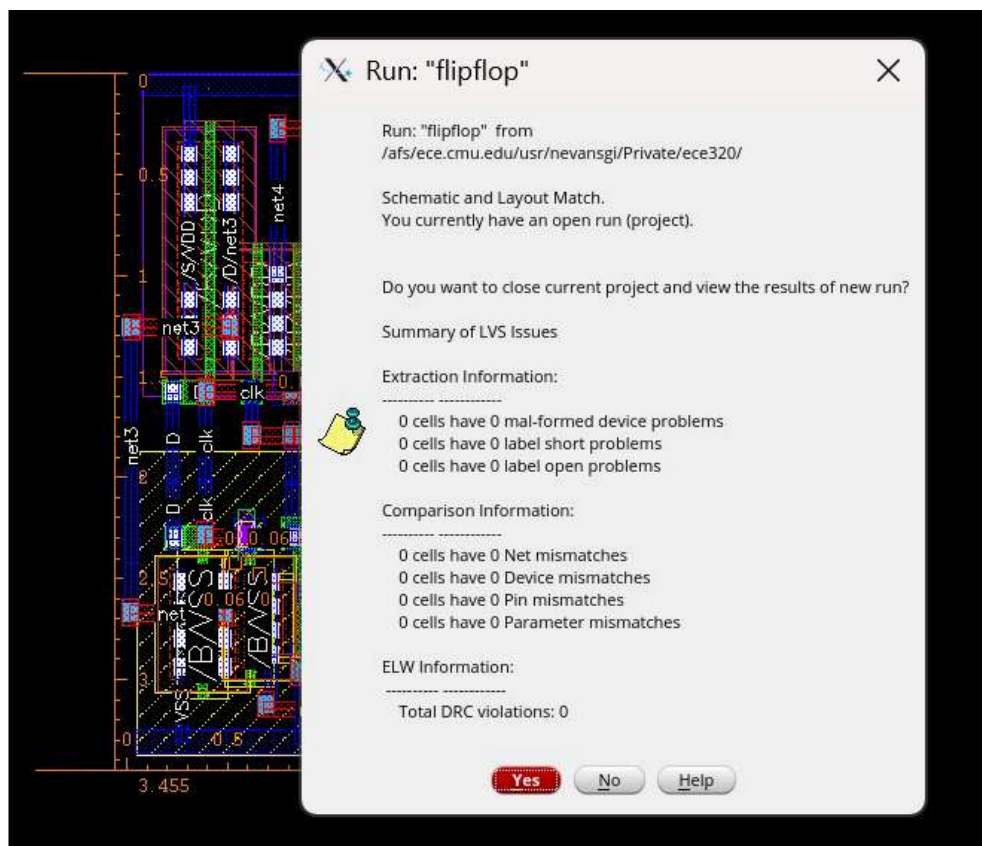
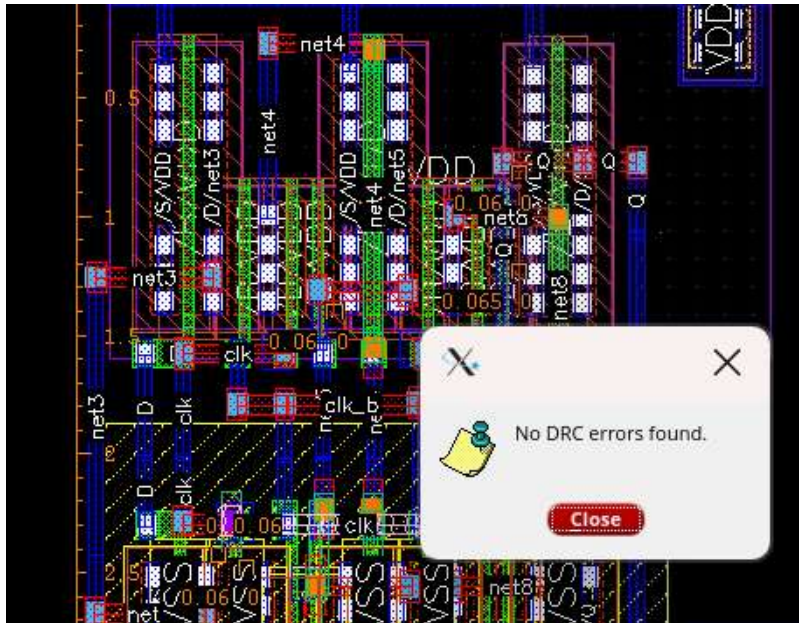
Module 3: Layout



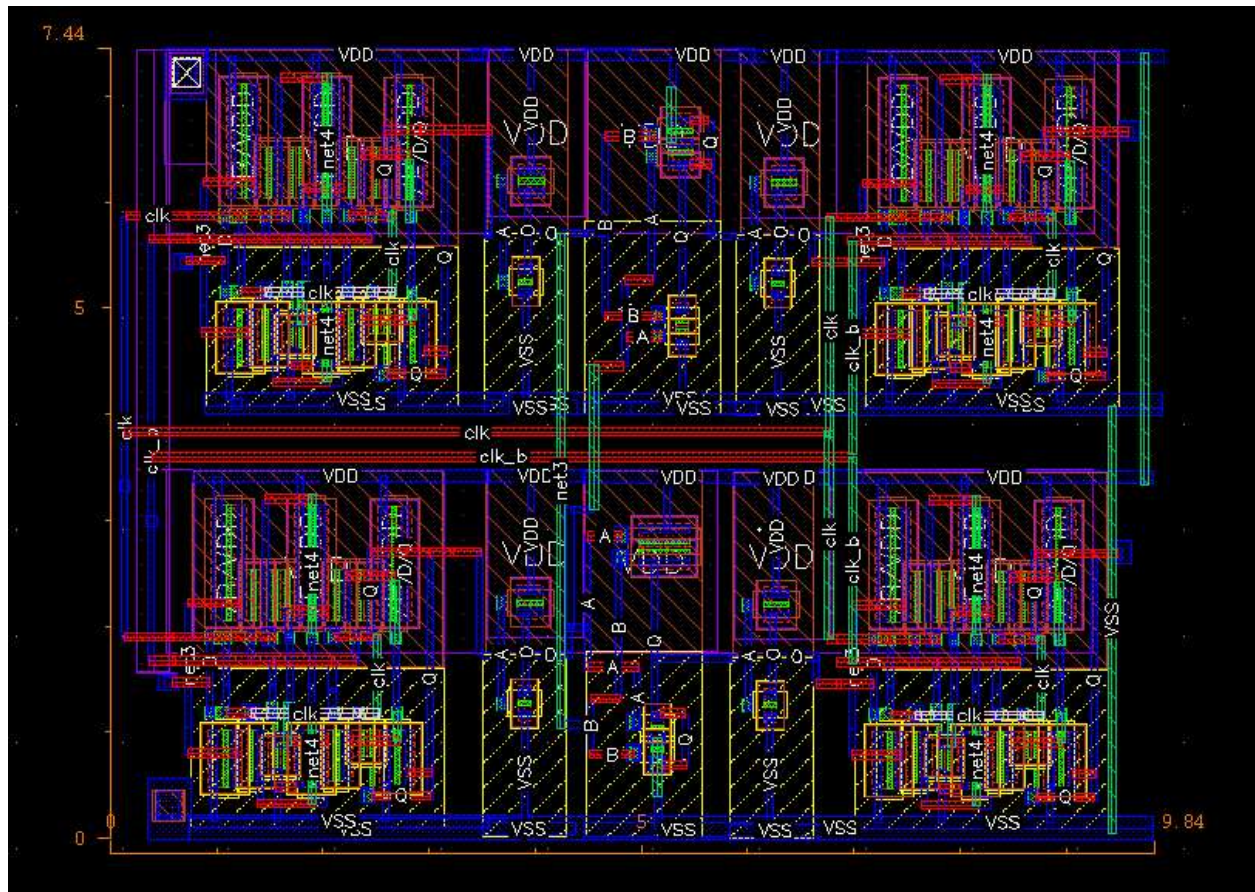
Total height (excluding taps): 3.455 μm

Total width (excluding taps): 2.465 μm

Total area: 8.517 μm^2



Full Layout



Total area: $73.21 \mu m^2$

Module 4: Parasitic Extraction

Calculating setup time post-extraction:

$$t_{CLK} - t_D = 1.438 \text{ ns} - 1.394 \text{ ps}$$

$$t_{SETUP} = 44 \text{ ps}$$

Setup slack post-extraction:

$$t_{COMB} = 85.4 \text{ ps}$$

$$t_{REQUIRED} = t_{CLK} - t_{SETUP} = 416.7 \text{ ps} - 44 \text{ ps} = 372.7 \text{ ps}$$

$$t_{SLACK} = t_{REQUIRED} - t_{COMB}$$

$$t_{SLACK} = 372.7 \text{ ps} - 85.4 \text{ ps} = 287.3 \text{ ps}$$

$$t_{SLACK} = 287.3 \text{ ps}$$